

A wave on a string is described by the equation $y(x, t) = 0.25 \sin(3.14x + 450t)$, where all numbers are in proper MKS units.

- (a) What is the frequency of the wave?
- (b) What is the period of the wave?
- (c) What is the wavelength of the wave?
- (d) What is the velocity of the wave?
- (e) What is the amplitude of the wave?
- (f) Is the wave travelling in the positive or negative x direction?
- (g) If the tension in the string is 100 N, what is the mass density, μ , of the string?

A flute should play an A ($f=440$ Hz) in its fundamental mode.

- (a) Sketch the node pattern of the flute in its fundamental mode.
- (b) What is the length of the flute's resonant cavity when sounding the A?
- (c) To sound "middle C" ($f=262$ Hz) the flute player closes holes on the flute making it longer. How long to sound middle C?
- (d) To sound a high A ($f=880$ Hz), the flute player cannot simply open more holes since the length will become too short? What do you do instead? Sketch the resulting node pattern.

The “A” string on a guitar is the second lowest string (E is the lowest). This one ideally plays an A ($f=110$ Hz) two octaves below the A from the previous problem. The length of the string is 0.648 meters and the tension is 100 N.

- (a) Sketch the node pattern when the string vibrates in its fundamental mode.
- (b) Find the mass per unit length, μ , of the string.
- (c) You wish to play a “C” ($f=131$ Hz) on this string. How do you accomplish this without changing the tension or μ of the string? Specify your answer as an exact numerical result.

Shown in the picture below is a saxophone.



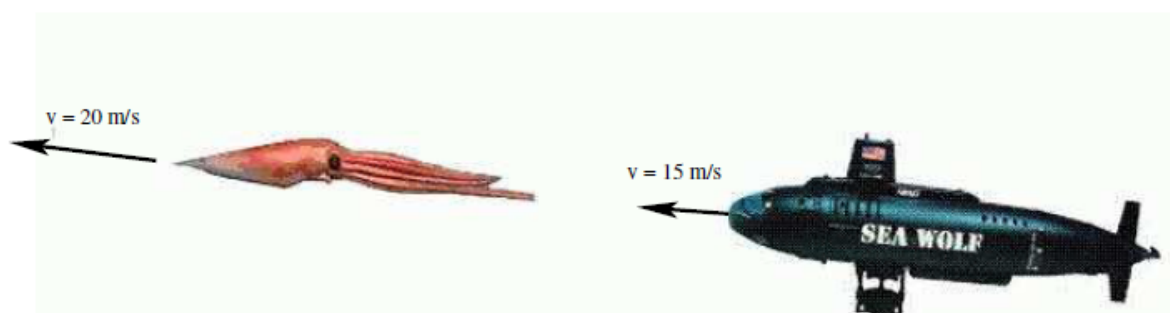
- (5 pts) Circle the sketch that represents the Node/Anti-node pattern when the saxophone is played in the low register in the fundamental mode. Label your circle with the letter "A".
- (5 pts) What length of the cavity, L , would produce a note with frequency $f = 116.5 \text{ Hz}$ (a B-flat).
- (5 pts) Circle the part of the saxophone that opens to make the saxophone play one octave higher (twice the previous frequency).
- (5 pts) Circle the node pattern when the saxophone is played as in part c). Label this circle as "D".

Two successive overtones of a closed organ pipe are 280 and 320 Hz.

- (a) What is the fundamental frequency?
- (b) What is the length of the organ pipe?
- (c) Prof H screwed up! This problem is flawed! Why?

You are standing 10 meters away from a jet plane and the sound level of 140 dB is painful. Where should you stand to reduce the sound level to 120 dB?

Shown in the figure below is a submarine chasing a squid. Luckily for the squid he is moving faster. The submarine emits a sonar pulse with a frequency of 12,000 Hz. Using 1450 m/s as the speed of sound in water



- (a) Frequency of sound heard by the squid.
- (b) Frequency of sound heard by the sub reflecting off the squid.
- (c) Beat frequency heard at the sub.